# Correlating two computations

Prediction and classification attacks between a reference computation and an encoded one

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*A tutorial: the general problem of correlating two computations, the three families of predictor that attack it, and what each of our protection plans does against each attack. Measurements are on the* $n$=128 *sliced sandwich with the Gray-code fold* (`docs/GRAY_FOLD_CG`).

## 1 The general setting

Take two computations run on the same inputs: $C$ (the reference — what we are trying to hide) and $C'$ (what an adversary holds). Fix a prefix $i$ of $C$ and a prefix $j$ of $C'$, run both on $s$ shared random inputs $x_1 \ldots x_s$, and you have two binary matrices

$$Y = C_i, \quad Y[a,b] = \big(\text{state of } C \text{ after } i \text{ gates on } x_a\big)_b \qquad (s \times n)$$

$$Z = C'_j, \quad Z[a,b] = \big(\text{state of } C' \text{ after } j \text{ gates on } x_a\big)_b \qquad (s \times m)$$

Every attack below tests whether $Z$ tells you about $Y$, evaluated at each $(i,j)$ and read as a heatmap. In our case $C' = G$, the gadgetized circuit, with $m = 4n$ wires against $C$'s $n$; the extra inputs are pinned to zero.

## 2 The trivial answer, and why security must be class-relative

Ask "are $Y$ and $Z$ dependent?" and the answer is **always yes**. Both are deterministic functions of the same $x$, and $G$ is an invertible encoding, so $Z$ determines $x$ determines $Y$. The mutual information is maximal by construction.

This dictates what can be measured. A test that is *consistent against all alternatives* — distance correlation, HSIC, any universal independence test — fires at full strength on every build, secure or not. **No test can certify "no dependence."** Every meaningful statement has the form

> *no adversary in class* **A** *can extract feature* **f** *of C's state from G's state, at sample budget s*,

so every number below is a **lower bound**, and the class must always be named. That is not modesty; it is the only claim the setting admits.

## 3 The unifying object: the cross-correlation spectrum

Fix $i, j$. For a target mask $\alpha \in \mathbb{F}_2^n$ and a predictor mask $\beta \in \mathbb{F}_2^m$, define

$$M[\alpha, \beta] \;=\; \mathbb{E}_x\Big[(-1)^{\,\alpha\cdot Y \,\oplus\, \beta\cdot Z}\Big].$$

This is the **Walsh cross-spectrum**. $|M| = 1$ means $\alpha \cdot Y$ is an exact affine function of $Z$; $M = 0$ means that parity carries nothing; in between are biases. Every instrument is a restriction of this one object:

| restriction | gives you |
|---|---|
| $\lvert M\rvert = 1$ exactly, $\beta$ unrestricted | **algebraic / exact** predictors (§4) |
| $\lvert M\rvert$ maximized, $\beta$ of bounded weight | **statistical** predictors (§5) |
| replace $\beta \cdot Z$ by any degree-$\leq D$ function of $Z$ | degree-$D$ versions of both |
| the full singular spectrum, over $\mathbb{R}$ | **canonical correlation** (§6) |

The matrix is $2^n \times 2^m$, so nobody enumerates it; every instrument picks a restriction, and its blind spots follow from that choice.

## 4 Algebraic (exact) predictors

**The question.** Is $\alpha \cdot Y$ *exactly* a degree-$\leq D$ function of $Z$, on every input?

**How it is answered.** Build the design matrix of all monomials of $Z$ up to degree $D$ — $N_D = \sum_{k \leq D} \binom{m}{k}$ columns — and test whether $\alpha \cdot Y$ lies in its GF(2) column span, with a held-out split to reject spurious fits. The answer is *binary per cell.*

**Our instrument.** `hmap_affine`, reporting $H(i, j)$ = mean over $C$'s $n$ target bits of the holdout error, scoring an inconsistent bit at exactly 0.5. So $H \approx 0$ is total leakage, $H \approx 0.5$ is hidden.

**Why it exists.** It is *invariant to the affine part of the encoding.* $G$ carries each value as a XOR of shares over unrelated wires; a raw Hamming comparison sees noise, and an early "the diagonal is destroyed" conclusion was exactly that artifact, corrected once the affine-invariant measure existed. Degree 2 additionally sees through any quadratic re-encoding.

**Its blind spot.** It cannot see *bias.* A relation holding on 99% of inputs scores the same as one holding on 50% — both are "not exact." This is why it and the statistical measures disagree, and why neither subsumes the other.

## 5 Statistical (bias) predictors

**The question.** How well can a *bounded* predictor **guess** $\alpha \cdot Y$ from $Z$?

**Our instrument.** `stress_battery`, with two families:

- **F1** — best over {constant, single wire, XOR of two wires}: $|\beta| \leq 2$, degree 1. A GF(2) analogue of correlation power analysis.
- **F2** — F1 plus one AND over a declared wire set (for us the 256 band wires): degree 2.

Three design points make the numbers readable:

1. **Lift, not agreement.** Each cell reports best − base rate, the base rate being the best *constant* predictor, so a merely skewed target scores zero.
2. **An overfit floor.** With $W = 512$ predictor wires the search is over ∼131k candidates, and maximizing over that many buys agreement by chance. The floor is the same search against a *random* target; `prominence` = raw − floor, and the leak gate is prominence $> 2 \cdot$ floor.
3. **Two gates.** `ALIGNED-LEAK` needs exploitable information *and* progress alignment ($\rho$, perm-$z$): the best-predicting $G$-prefix must advance as the $C$-prefix does.

**Scaling.** The floor behaves as $\sqrt{\ln N/2s}$, so resolving a bias $\delta$ needs $s \gtrsim \ln N/\delta^2$, while

$$\text{cost} \;=\; \text{rows} \times \text{cols} \times n_t \times \left(\tfrac{W^2}{2} + 2A^2\right) \times \tfrac{s}{64}$$

is *linear* in $s$ and *quadratic* in wires. Resolving a leak $k\times$ smaller costs $k^2$; widening $G$ from $4n$ to $8n$ needs only ∼12% more samples but ∼4.5× the time.

# 6 Canonical correlation, and the right basis

Classical CCA maximizes $\mathrm{corr}(Ya, Zb)$ over *real* vectors $a, b$, solved by an SVD of the whitened cross-covariance; the singular values are the canonical correlations. The translation to our setting is the useful part. Encode bits as ±1; then

$$\text{XOR of a set of bits} \;\longleftrightarrow\; \text{PRODUCT of the } \pm 1 \text{ variables} \;\longleftrightarrow\; \text{one Fourier character.}$$

So a GF(2)-linear relation is a *multiplicative* monomial, not an additive one. Consequently:

- **Real-linear CCA** on raw bits finds *additive* combinations — essentially Hamming-weight relations. That is the classical side-channel leakage model, natural when the observable is analog.
- **GF(2) character analysis** finds XOR relations. That is *linear cryptanalysis*, natural for a Boolean circuit.

Both are canonical correlation over different bases. For our problem the XOR basis is the right one, and real CCA on raw wire values would mostly measure Hamming-weight structure the encoding neither especially protects nor exposes. The GF(2) analogue of CCA's full singular spectrum is the entire Walsh cross-spectrum of §3 — which is why practical instruments are all *restrictions* rather than decompositions.

# 7 Our case: $G$ is a masking scheme, so order matters

The product-share encoding represents each value as

$$v \;=\; w \,\oplus\, A_1 \,\oplus\, \dots \,\oplus\, A_k, \qquad A_r = \text{product of } d_r \text{ band bits},$$

with $w$ the carrier. This is **Boolean masking**, and the side-channel literature on masked implementations transfers directly.

**The piling-up lemma.** For a degree-$d$ product of uniform independent bits, $\mathbb{E}[(-1)^A] = 1 - 2^{1-d}$: a degree-2 atom contributes 0.5, a degree-3 atom 0.75. If the atoms are variable-disjoint (which `draw_slot` enforces per value) the biases multiply:

$$\varepsilon \;=\; \prod_r \big(1 - 2^{1-d_r}\big), \qquad \mathrm{corr}(\mathrm{carrier}, \mathrm{value}) = \varepsilon,$$

so a predictor reading the bare carrier guesses the value with agreement $(1+\varepsilon)/2$. $\varepsilon$ **is the single number governing the statistical attack.**

**Order of attack.** To *remove* an atom of degree $d$ the adversary must compute a degree-$d$ monomial of band wires — a $d$-th order combining function. After cancelling a set $S$ the residual is $\prod_{r \notin S}$. Hence F1 sees the full $\varepsilon$; F2 can cancel one degree-2 atom and see $\varepsilon/0.5 = 2\varepsilon$; an F3 would cancel a degree-3 atom and see $\varepsilon/0.75$.

**The design tension, in one line.** A degree-2 atom is a *stronger* statistical masker (0.5 vs 0.75) but a *weaker* algebraic one — it lies inside a degree-2 adversary's span, where a degree-3 atom does not. Low degree buys statistics; high degree buys algebra.

# 8 Results: each plan against each attack

Five mask plans, all with the Gray fold, $n$=128, same source $C$ and sandwich, $s$ = 8192, plate geometry matched per build.

## 8.1 Predicted bias, before and after cancellation

| plan | atoms | $\varepsilon$ (order 1) | after one 2-AND | after one 3-AND |
|---|---|---:|---:|---:|
| `[2,3,3]` *(shipped)* | d2, d3, d3 | 0.28125 | 0.5625 | 0.375 |
| `[2,2,3]` | d2, d2, d3 | 0.1875 | 0.375 | 0.25 |
| `[2,2,3,3]` | d2, d2, d3, d3 | 0.140625 | 0.28125 | 0.1875 |
| `[2,2,2,3]` | d2×3, d3 | 0.09375 | 0.1875 | 0.125 |
| `[2,2,2,3,3]` | d2×3, d3, d3 | 0.0703 | 0.140625 | 0.09375 |

## 8.2 Measured

| plan | $\varepsilon$ | gates | vs base | reachable | F1 raw | F2 raw | verdict F1/F2 |
|---|---:|---:|---:|---:|---:|---:|---|
| `[2,3,3]` | 0.28125 | 808,618 | — | 95.47% | 0.0817 | 0.1111 | **LEAK / LEAK** |
| `[2,2,3]` | 0.1875 | 634,790 | **−21%** | 97.01% | 0.0536 | 0.0782 | flat / **LEAK** |
| `[2,2,3,3]` | 0.1406 | 864,497 | +6.9% | 96.16% | 0.0435 | 0.0660 | flat / flat |
| `[2,2,2,3]` | 0.09375 | 692,653 | **−14%** | **97.53%** | 0.0318 | 0.0487 | flat / flat |
| `[2,2,2,3,3]` | 0.0703 | 924,284 | +14% | 96.56% | 0.0258 | 0.0409 | flat / flat |

"Raw" = prominence + floor; compare arms by raw, because the floor is a noisy single-sample estimate (§9).

## 8.3 The law

Regressing the measured signal on the predicted piling-up bias:

$$\mathrm{F1}_{\mathrm{raw}} = 0.262\,\varepsilon + 0.007 \quad (R^2 = 0.996), \qquad \mathrm{F2}_{\mathrm{raw}} = 0.330\,\varepsilon + 0.018 \quad (R^2 = 0.998).$$

Five plans spanning a 4× range of $\varepsilon$, and the statistical leak is **linear in the piling-up product** to within half a percent. This identifies the leak as the carrier's *marginal mask bias* and nothing else.

Two readings follow. The **slope ratio** $0.330/0.262 = 1.26$ is the payoff of second order: theory says a *perfectly targeted* 2-AND doubles the effective bias (ratio 2.0), so the blind search over $\binom{256}{2}$ pairs realizes about a quarter of it — a targeted F2 using ledger knowledge would reach the full factor. The **intercepts** (0.007, 0.018) are residual overfit inflation the floor subtraction does not remove, a reminder that "flat" is a statement about a threshold, not about zero.

## 8.4 What the algebraic attack says, and why it is flat across the board

`hmap_affine`, matched geometry, reads every plate measured as dead: depthMed = 0.0000, depth 0.044–0.046 against an "alive" calibration of ≈0.35, and **zero genuine interior rows**.

| plate | degree | depth | depthMed | $\rho$ | interior leaks |
|---|---|---|---|---|---|
| wide fold `[2,3,3]` | 1 | 0.0438 | 0.0000 | 0.017 | 0 / 88 |
| Gray fold `[2,3,3]` | 1 | 0.0436 | 0.0000 | 0.016 | 0 / 88 |
| wide fold `[2,3,3]` | 2 | 0.0455 | 0.0000 | −0.042 | 0 / 88 |
| Gray fold `[2,3,3]` | 2 | 0.0439 | 0.0000 | 0.016 | 0 / 88 |
| `[2,2,3]` | 2 | 0.0439 | 0.0000 | 0.016 | **0 / 88** |
| `[2,2,2,3]` | 2 | 0.0444 | 0.0000 | −0.047 | **0 / 88** |

The last two rows are the ones that mattered: `[2,2,3]` and `[2,2,2,3]` trade degree-3 atoms for degree-2 ones, so they are where a degree-2 exact adversary would surface first if the margin were thinner than theory says. It is not. (`[2,2,3,3]` and `[2,2,2,3,3]` were not run at degree 2; they carry *more* degree-3 material than the plans that passed, so they are covered a fortiori.)

That is forced, not lucky. Exact degree-$D$ reconstruction requires *every* atom to lie in the degree-$D$ span, i.e. $D \geq \max_r d_r$. All five plans keep at least one degree-3 atom, so **the algebraic verdict is identical for every plan** and they differ *only* statistically.

What does differ is **redundancy, not threshold**: `[2,2,3]` and `[2,2,2,3]` hold a single degree-3 atom, so if one is ever compromised the value drops into degree-2 range; the other three plans keep two. A pure `[2,2,2]` plan would have no algebraic margin at all — statistically the best of all, and dead at $D = 2$.

## 8.5 The Pareto surprise

**`[2,2,2,3]` dominates the shipped `[2,3,3]` on every axis measured at once**: 14% fewer gates, higher store-reachability (97.53% vs 95.47%), 3× lower statistical leak, identical algebraic standing.

It is affordable because of the Gray fold. Under the wide fold a block emits $(1+k)^{\text{arity}}$ fragments, so a mask term is *multiplicative* — 3 to 4 terms costs +56%. Under the Gray fold the product part is a fixed ∼9 gates regardless of $k$, and a term costs only its own gather (∼1 gate at degree 2, ∼4 at degree 3). Mask-plan cost becomes **additive**, and trading a degree-3 atom for degree-2 atoms makes the circuit *smaller*.

## 9 How to read these numbers without fooling yourself

Five traps, all of which bit during this work.

1. **Never read a heatmap by its mean** — it saturates near 0.5. Read the ridge.

2. **Never read $\rho$ alone on a plate with ports.** Both axes have unencoded ends; cell $(0,0)$ compares $C$'s input against $G$'s input wires and reads perfectly in *every* build. On such a plate $\rho$ is two plateaus ranked against noise in a dead middle: two equally-dead artefacts once scored $\rho = 0.448$ and 0.019. Report **interior rows with prominence**. Our degree-1 plates report perm-$z = 4.91$, $p = 0.0003$ — and are dead.

3. **The overfit floor is a noisy single-sample estimate, and the verdict rides on it.** Measured floors are not monotone in $s$: 0.0454, 0.0205, 0.0242, 0.0136 at $s = 2048, 4096, 8192, 16384$. The *same circuit* scored `flat` at $s$=2048 and `ALIGNED-LEAK` at $s$=4096. Compare by raw.

4. **A fixed `--g-step` across circuits of different length is a confound.** cols = $|G|$/g_step and row-best is a max over columns, so a longer circuit scores leakier for free. A +24%-gate arm ran 52 columns against 42 and looked 13% worse; matching geometry removed a third to a half of that.

5. **"Flat" means "below threshold at this $s$."** Floor $\propto 1/\sqrt{s}$. `[2,2,2,3]`'s F1 raw of 0.0318 sits just above a 0.0239 floor with $\rho = 0.996$ — the diagonal is still there; it no longer clears the bar.

## 10 What we did not test

- **A structure-aware adversary.** The mask atoms are *readable off the gate list* — the fold's own gates name the band wires of each atom. Such an adversary does not search; it reads and cancels exactly, achieving the full order-$d$ factor rather than the blind search's quarter. Strictly stronger than anything run here, and what rolling, re-sourcing and epochs exist to complicate.
- **Targeted F3**, cancelling a degree-3 atom ($\varepsilon/0.75$): trivial targeted, $\sim$2800$\times$ F2's cost blind.
- **Differential correlation**, $\Delta C$ against $\Delta G$ under input differences. The battery's `traj` target is a *temporal* difference, not an input one.
- **Machine-learned predictors** over all $4n$ wires.
- **Universal independence tests** are deliberately absent; §2 says why.

## 11 Practical summary

- The statistical leak is the carrier's marginal mask bias, **linear in the piling-up product** ($R^2 \approx 0.996$). To reduce it, reduce $\varepsilon$.
- Refresh **rate** and refresh **source** do not touch it — measured flat across `--prod-refill-data` 0/50/100. They change the band's provenance, not its marginal, and the marginal is what F1 sees.
- Low-degree-heavy plans win statistically at *every* attack order; degree-3 atoms buy the algebraic margin. Keep at least two for redundancy.

- `[2,2,2,3]` is Pareto-better than the shipped `[2,3,3]` on size, digestibility and statistical leak, with identical algebraic standing but one fewer degree-3 atom of margin. `[2,2,2,3,3]` buys that margin back for +14% gates and the lowest leak measured.